RONDE&SCHWARZ
Make ideas real

Homework 4:

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Individual Values:

$A_4 = 355$ $B_4 = 9920$ $C_4 = 7700$ $P_4 = 7380$

$PP_1: N_1 = 9920 + 180 P_1 + 0{,}02 P_1^2$

$PP_2: N_2 = 7700 + 185 P_2 + 0{,}04 P_2^2$

$PP_3: N_3 = 7380 + 190 P_3 + 0{,}03 P_3^2$

The total load Power is 355 MW, it means

$$355 - P_1 - P_2 - P_3 = 0$$

and Objective function is:

$f = N_1 + N_2 + N_3 \Rightarrow$

$$f = 9920 + 180 P_1 + 0{,}02 P_1^2 + 7700 + 185 P_2 + 0{,}04 P_2^2 + 7380 + 190 P_3 + 0{,}03 P_3^2$$

Then Lagrange function will be:

$$F = 9920 + 180P_1 + 0{,}02P_1^2 + 7700 + 185P_2 + 0{,}04P_2^2 + 7380 + 190P_3 + 0{,}03P_3^2 + \mu(355 - P_1 - P_2 - P_3)$$

$$\mu = \frac{\partial F}{\partial P_1} = 180 + 0{,}04P_1 \rightarrow$$

$$\mu = \frac{\partial F}{\partial P_2} = 185 + 0{,}08P_2 \rightarrow$$

$$\mu = \frac{\partial F}{\partial P_3} = 190 + 0{,}06P_3 \rightarrow$$

We got:

$$180 + 0{,}04P_1 = 185 + 0{,}08P_2 = 190 + 0{,}06P_3$$

and $$355 - P_1 - P_2 - P_3 = 0$$

from these equations we can get values:

$180 + 0{,}04\,P_1 = 185 + 0{,}08\,P_2$ | $180 + 0{,}04\,P_1 = 190 + 0{,}06\,P_3$

$P_2 = -62{,}5 + 0{,}5\,P_1$ | $P_3 = -166{,}67 + 0{,}67\,P_1$

We can substitute:

$$355 - P_1 + 62{,}5 - 0{,}5\,P_1 + 266{,}67 - 0{,}67\,P_1 = 0$$

$$584{,}17 = 2{,}17\,P_1$$

$P_1 = 269{,}2$ obviously it is higher than load,

we have to give fixed value to P_1,

Then, $\boxed{P_1 = 175\ MW}$

Then P_2 and P_3 will be:

$$355 - 175 - P_2 - P_3 = 0$$

$$180 - P_2 - P_3 = 0$$

$$\begin{cases} 185 + 0{,}10\,\delta P_2 = 190 + 0{,}06\,P_3 \\ 180 - P_2 - P_3 = 0 \end{cases}$$

$$P_2 = 62{,}5 + 0{,}175\,P_3$$

$$62{,}5 + 0{,}175\,P_3 + P_3 = 180$$

$$\boxed{\begin{aligned} P_3 &= 67{,}14 \text{ MW} \\ P_2 &= 112{,}855 \text{ MW} \end{aligned}}$$

Then final Power dissodence is:

$$P_1 = 175 \text{ MW}$$

$$P_2 = 112{,}86 \text{ MW}$$

$$P_3 = 67{,}14 \text{ MW}$$

Total Cost: $N_1 + N_2 + N_3 = 91392{,}93$ CZK/h2